

Tide retrospective: flat perturbations are invisible (germbij Proposition 4.1)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The germbij note’s identifiability programme proved the positive half of its story: analytic potentials are pinned by their expectation values, because an analytic pencil difference obeys a sector lower bound κt^γ that contradicts superpolynomial decay. The note’s Proposition 4.1 is the negative half, and the reason the analyticity hypothesis is not decoration: a *flat* perturbation — smooth, nonnegative, vanishing to infinite order at the minimum — changes every observable integral by less than any power of $1/t$. Without it, the identifiability theorem reads as a technical convenience; with it, analyticity is exposed as the exact boundary of what expectation values can see.

This was scoped as a candidate multi-tide programme, but scoping collapsed it to a single tide: no expansion machinery is needed at all. The pointwise identity

$$e^{-tL} - e^{-t(L+f)} = e^{-tL} (1 - e^{-tf}), \quad 0 \leq 1 - e^{-s} \leq s,$$

together with a *global* polynomial domination of f , reduces the whole difference to one closed-form Gaussian moment that the seabed already owns.

2 The thing formalised

Two declarations in `Laplace/OneD/FlatInvisible.lean`, sorry-free with zero warnings. The main theorem¹: if f is continuous with $0 \leq f \leq M$ and flat at 0 (for every n there are $C \geq 0, \delta > 0$ with $f(x) \leq Cx^{2n}$ on $|x| \leq \delta$), then for every continuous compactly supported φ and every N there are $K \geq 0$ and $T \geq 1$ with

$$\left| \int \varphi e^{-tx^2/2} dx - \int \varphi e^{-t(x^2/2+f)} dx \right| \leq \frac{K}{t^N} \quad (t \geq T).$$

The corollary² packages this as `IsLittleO`: the difference is $o(t^{-N})$ for every N — precisely the hypothesis shape that `lower_bound_not_superpolynomial` in `Laplace/Decay.lean` refutes for analytic pencil differences. The two halves of the note’s story now meet in the same vocabulary: analytic differences *cannot* decay superpolynomially, flat ones *do*.

¹`flat_perturbation_invisible.`

²`flat_perturbation_superpolynomial.`

3 Proof strategy

Three moves, none requiring new infrastructure. First, a sup bound M_φ for $|\varphi|$ from compact support. Second, the global domination: near 0 flatness gives $f \leq Cx^{2N+2}$; away from 0 boundedness gives $f \leq M \leq (M/\delta^{2N+2})x^{2N+2}$, so $f \leq (C + M/\delta^{2N+2})x^{2N+2}$ everywhere. Third, the pointwise bound

$$|\varphi e^{-tx^2/2} - \varphi e^{-t(x^2/2+f)}| \leq M_\varphi e^{-tx^2/2} tf \leq M_\varphi(C+D)tx^{2N+2}e^{-tx^2/2},$$

integrated by the seabed’s closed-form moment (`integral_pow_mul_exp_neg_t_sq_half`), which yields $c_N t \cdot t^{-(N+3/2)} \leq c_N t^{-N}$ for $t \geq 1$. The numerical check (run *before* the consult, per the honesty rule) confirmed superpolynomial decay for $f = e^{-1/x^2}$: $|\Delta(t)|$ at $t = 20, 80, 320$ is $8.4 \times 10^{-3}, 5.1 \times 10^{-5}, 4.2 \times 10^{-10}$, with $\Delta \cdot t^3 \rightarrow 0$ and $\Delta \cdot t^5$ collapsing after onset.

4 Roadblocks and resolutions

One genuinely new failure mode. The double-factorial notation $n!!$ needs `open scoped Nat`, but that scope also defines φ as notation for `Nat.totient` — so the theorem binder $\{f \varphi : \mathbb{R} \rightarrow \mathbb{R}\}$ stopped parsing (*unexpected token, expected* $\}$) the moment the open was added. Identifiers merely containing φ survive; only the bare binder dies. Resolution: skip the scoped open and write `Nat.doubleFactorial` explicitly; the moment lemmas stated with the notation still apply since it denotes the same constant. Promoted to the repository error catalogue.

The only other friction was two known-catalogue items (the `le_or_lt` rename, and a guessed integrability lemma name — the seabed’s actual lemma is `integrable_pow_mul_exp_neg_mul_sq` at $c = t/2$, bridged by a one-line `ring` congruence). The `IsLittleO` corollary, including its `rpow/npow` bookkeeping, built on the first attempt via `isLittleO_iff_tendsto`’ and the ratio $t^{-(N+1)}/t^{-N} = t^{-1}$.

5 The deliberation

GPT-5.6 Sol endorsed the candidate exactly as drafted: the even-power flatness hypothesis (composes with the Gaussian moment lemmas without `rpow` leakage), $T = 1$, the file name, and the all-orders K/t^N form as the quantitative core. Its one substantive addition was adopted:

The all-orders $O(t^{-N})$ statement faithfully captures “no asymptotic expansion coefficient changes.” [...] I would make this theorem the elementary quantitative result, and a short corollary translating it into `Laplace.Ddecay`’s superpolynomial vocabulary. That keeps the analytic proof independent of the packaging.

Its suggested helper decomposition was declined (each helper is used once; the monolithic proof is ~ 260 lines and readable), and its recommendation to defer the classic witness $f = e^{-1/x^2}$ to a separate declaration was accepted — GPT supplied the flatness route ($s^n/n! \leq e^s$ gives $e^{-1/x^2} \leq n!x^{2n}$ globally, so $C = n!$ and any δ works), which makes that follow-up tide well-scoped if commissioned.

6 Mathlib vs new

Everything analytic was Mathlib or seabed lookup: the compact-support sup bound, compact-support integrability, the inequality $1 + s \leq e^s$, the seabed’s Gaussian moment and integrability

lemmas, and the tendsto characterisation of IsLittleO.³ New content: the statement itself, the global domination argument, and the assembly.

7 What the next tide inherits

The natural follow-up is the witness declaration: a concrete flat $f = e^{-1/x^2}$ instance satisfying every hypothesis of `flat_perturbation_invisible`, with flatness from the factorial bound above (Mathlib’s `expNegInvGlue` is the reference point for continuity at 0). With the witness in place the note’s Proposition 4.1 is fully witnessed rather than conditionally quantified. Beyond that, the remaining germbij commissions are the higher cumulant rungs of Theorem 3.1 and the multivariate §7.4 programme.

³Respectively `HasCompactSupport.exists_bound_of_continuousOn`, `Continuous.integrable_of_hasCompactSupport`, `Real.add_one_le_exp`, `integral_pow_mul_exp_neg_t_sq_half` and `integrable_pow_mul_exp_neg_mul_sq`, and `isLittle0_iff_tendsto`.